

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 02 · Quadratics

Questions

Student book · questions with writing space

- 44 pages
- Past papers up to June 2026

Quadratics

Your classified route

15
questions

105
marks

5
families

10
variants

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CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Quadratics

Families and variants

F01 • Quadratic Equation Methods

V01 Direct Exact Quadratic Solution

V02 Square-root or Fractional-power Substitution

V03 Even-power or Biquadratic Substitution

V04 Odd-power Quadratic Substitution

F02 • Completed-square Form and Parabola Features

V01 Complete the Square, Read and Sketch

V02 Complete the Square, then Intersect and Define a Region

F03 • Construct a Quadratic from Graph Data

V01 Roots, Symmetry and a Point

V02 Vertex Form and a Point

F04 • Discriminant and Root-count Conditions

V01 Parameter Range from a Required Root Count

V02 Curve Tangency or Separation

F05 • Quadratic Modelling from Constraints

V01 Sum-product Geometric Constraints

Quadratic Equation Methods

WHAT THIS FAMILY ASKS

Solve a quadratic directly or reveal one by substituting for a repeated expression, then map all admissible roots back to the original variable.

VARIANT ROUTE

V02 Square-root or Fractional-power Substitution

V03 Even-power or Biquadratic Substitution

3 Complete Classified questions

19 total marks

8. Solve, using algebra, the equation

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

Fully simplify your answers, writing them in the form $a + b\sqrt{c}$, where a , b and c are integers to be found.

(5)

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5. (a) Find, using algebra, all solutions of

$$20x^3 - 50x^2 - 30x = 0$$

(3)

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$$20x^3 - 50x^2 - 30x = 0$$

(b) Hence find all real solutions of

$$20(y + 3)^{\frac{3}{2}} - 50(y + 3) - 30(y + 3)^{\frac{1}{2}} = 0$$

(4)

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4.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

(i) Using the laws of indices, solve

$$3^{2p-5} = \frac{1}{9}\sqrt{27}$$

(3)

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(ii) Find the real roots of the equation

$$x^2 + 1 = \frac{36}{x^2 - 4}$$

(4)

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Completed-square Form and Parabola Features

WHAT THIS FAMILY ASKS

Convert a quadratic to completed-square form, read its turning point and symmetry, and use these features in sketches, intersections or region descriptions.

VARIANT ROUTE

V01 Complete the Square, Read and Sketch

V02 Complete the Square, then Intersect and Define a Region

5 Complete Classified questions

39 total marks

2.

$$f(x) = 11 - 4x - 2x^2$$

(a) Express $f(x)$ in the form

$$a + b(x + c)^2$$

where a , b and c are integers to be found.

(3)

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$$f(x) = 11 - 4x - 2x^2$$

(2)

2.

$$f(x) = 11 - 4x - 2x^2$$

(c) Write down the equation of the line of symmetry of C .

(1)

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3. (a) Express $3x^2 + 12x + 13$ in the form

$$a(x + b)^2 + c$$

where a , b and c are integers to be found.

(3)

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$$a(x+b)^2+c$$

(b) Hence sketch the curve with equation $y = 3x^2 + 12x + 13$

On your sketch show clearly

- the coordinates of the y intercept
- the coordinates of the turning point of the curve

(3)

Figure 1 shows a Cartesian coordinate system with a shaded region R . The region is bounded by a curve l and a line C . The line C passes through the point $(-1, 28)$ and intersects the x -axis at point P . The origin is labeled O .

Figure 5 shows part of the curve C with equation $y = f(x)$ where

(a) Write $2x^2 - 12x + 14$ in the form

where a , b and c are constants to be found.

Figure 5 shows part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^2 - 12x + 14$$

Given that C has a minimum at the point P

(b) state the coordinates of P

(1)

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Figure 5 shows part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^2 - 12x + 14$$

The line l intersects C at $(-1, 28)$ and at P as shown in Figure 5.

- (c) Find the equation of l giving your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)


$$f(x) = 2x^2 - 12x + 14$$

(d) Use inequalities to define the region R .

(3)

The figure shows a Cartesian coordinate system with x and y axes. The origin is labeled O. A curve, labeled C, is plotted in the first quadrant. A straight line, labeled l, is also plotted. The line l is tangent to the curve C at a point labeled M. The curve C and the line l intersect at another point labeled P. The region bounded by the y-axis, the curve C, and the line l is shaded and labeled R.

In this question you must show all stages of your working. Solutions relying on calculator technology are not acceptable.

- $$a(x + b)^2 + c$$

(3)

Figure 1 shows a sketch of the curve C with equation $y = 2x^2 - 16x + 50$

Given that in Figure 1

- M is the minimum point on C
- line l passes through the origin and intersects C at the points M and P

(b) find, using algebra and showing your working, the x coordinate of P .

(5)



- M is the minimum point on C
- line l passes through the origin and intersects C at the points M and P

(c) Use inequalities to fully define R .

5. The curve C has equation

$$y = -2x^2 + 12x + 7$$

(a) Write $-2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$, where a , b and c are constants.

(3)

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$$y = -2x^2 + 12x + 7$$

The point P is the maximum point on C .

(b) Find the coordinates of P .

(2)

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$$y = -2x^2 + 12x + 7$$

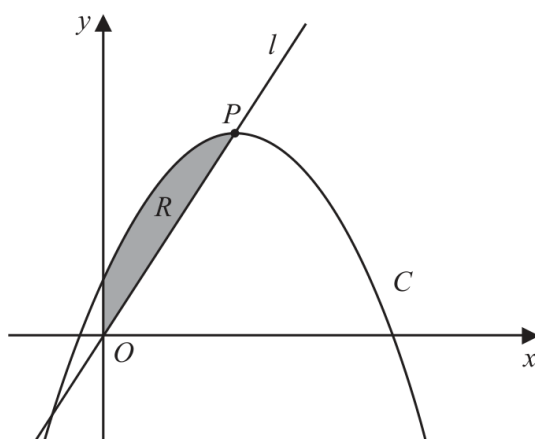


Figure 1

Figure 1 shows a sketch of the curve C and a line l .

The line l passes through the point P and the origin O .

The region R , shown shaded in Figure 1, is bounded by C , l and the y -axis.

(c) Use inequalities to define R .

(2)

Construct a Quadratic from Graph Data

WHAT THIS FAMILY ASKS

Select a root form or vertex form from graph information and determine its scale using another point or turning value.

VARIANT ROUTE

V01 Roots, Symmetry and a Point

V02 Vertex Form and a Point

2 Complete Classified questions

18 total marks

Figure 2 shows a sketch of the straight line l and the curve C .

(a) find an equation for l , writing your answer in the form $y = mx + c$, where m and c are constants to be found.

Given that C

- has equation $y = f(x)$ where $f(x)$ is a quadratic expression
- has a minimum point at $(7, -18)$
- cuts the x -axis at 4 and at k , where k is a constant

(b) deduce the value of k ,

(1)

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Given that C

- has equation $y = f(x)$ where $f(x)$ is a quadratic expression
- has a minimum point at $(7, -18)$
- cuts the x -axis at 4 and at k , where k is a constant

(c) find $f(x)$.

(3)

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The region R is shown shaded in Figure 2.

(d) Use inequalities to define R .

(2)

9. The curve C_1 has equation $y = f(x)$.

Given that

- $f(x)$ is a quadratic expression
- C_1 has a maximum turning point at $(2, 20)$
- C_1 passes through the origin

(a) sketch a graph of C_1 showing the coordinates of any points where C_1 cuts the coordinate axes,

(2)

9. The curve C_1 has equation $y = f(x)$.

Given that

- $f(x)$ is a quadratic expression
- C_1 has a maximum turning point at $(2, 20)$
- C_1 passes through the origin

(b) find an expression for $f(x)$.

(3)

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9. The curve C_1 has equation $y = f(x)$.

Given that

- $f(x)$ is a quadratic expression
- C_1 has a maximum turning point at $(2, 20)$
- C_1 passes through the origin

The curve C_2 has equation $y = x(x^2 - 4)$

Curve C_1 and C_2 meet at the origin, and at the points P and Q

Given that the x coordinate of the point P is negative,

(c) using algebra and showing all stages of your working, find the coordinates of P

(5)

Discriminant and Root-count Conditions

WHAT THIS FAMILY ASKS

Translate a required number of real roots or curve intersections into a sign condition on the discriminant, then solve the resulting parameter equation or inequality.

VARIANT ROUTE

V01 Parameter Range from a Required Root Count

V02 Curve Tangency or Separation

4 Complete Classified questions

23 total marks

4. Given that the equation

$$kx^2 + 6kx + 5 = 0 \quad \text{where } k \text{ is a non zero constant}$$

has no real roots, find the range of possible values for k .

(4)

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6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The equation

$$4(p - 2x) = \frac{12 + 15p}{x + p} \quad x \neq -p$$

where p is a constant, has two distinct real roots.

(a) Show that

$$3p^2 - 10p - 8 > 0$$

(3)

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$$3p^2 - 10p - 8 > 0$$

(b) Hence, using algebra, find the range of possible values of p

(3)

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10.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

$$(k - 1)x^6 + 4x^3 + (k - 4) = 0 \quad \text{where } k \text{ is a constant}$$

(a) Find the exact solutions to the given equation for $k = 4.5$

(3)

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10.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

$$(k-1)x^6 + 4x^3 + (k-4) = 0 \quad \text{where } k \text{ is a constant}$$

(b) Find the set of possible values of k for which the given equation has no real roots.

(4)

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4. The curve C_1 has equation

$$y = x^2 + kx - 9$$

and the curve C_2 has equation

$$y = -3x^2 - 5x + k$$

where k is a constant.

Given that C_1 and C_2 meet at a single point P

(a) show that

$$k^2 + 26k + 169 = 0$$

(3)

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The curve C_1 has equation

$$y = x^2 + kx - 9$$

and the curve C_2 has equation

$$y = -3x^2 - 5x + k$$

where k is a constant.

Given that C_1 and C_2 meet at a single point P

$$k^2 + 26k + 169 = 0$$

(b) Hence find the coordinates of P

(3)

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Quadratic Modelling from Constraints

WHAT THIS FAMILY ASKS

Translate linked real-world constraints into a quadratic equation and select the solution satisfying the original meaning.

VARIANT ROUTE

V01 Sum-product Geometric Constraints

1 Complete Classified questions

6 total marks

2.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A rectangular sports pitch has length x metres and width y metres, where $x > y$

Given that the perimeter of the pitch is 350 m,

(a) write down an equation linking x and y

(1)

2.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A rectangular sports pitch has length x metres and width y metres, where $x > y$

Given that the perimeter of the pitch is 350 m,

Given also that the area of the pitch is 7350 m^2

(b) write down a second equation linking x and y

(1)

2.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A rectangular sports pitch has length x metres and width y metres, where $x > y$

Given that the perimeter of the pitch is 350 m,

(c) hence find the value of x and the value of y

(4)